

P-FTNet: Pilot-Conditioned and Feature-Augmented Transformer Network for Equalization Under Extreme Delay Spread

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Abstract—In dense urban environments, severe multipath propagation and high delay spread create long-range temporal dependencies that conventional equalizers and existing deep learning models often struggle to capture. To address these challenges, we propose a Pilot-aided Feature-augmented Transformer Network (P-FTNet) for robust equalization under such extreme conditions. By integrating a pilot-aided module with multidimensional feature extraction module, the proposed architecture explicitly leverages reference information to enhance global sequence modeling. Experimental results demonstrate that P-FTNet significantly outperforms state-of-the-art methods, particularly in pilot-constrained scenarios, while exhibiting exceptional robustness to variations channel conditions.

Index Terms—channel equalization, high delay spread, pilot-aided, deep learning, transformer

I. INTRODUCTION

HIGH-SPEED single-carrier communications in urban scenarios face severe temporal dispersion due to dense multipath propagation. The resulting deep inter-symbol interference (ISI) spans multiple symbol durations as the delay spread exceeds the symbol interval. Accompanied by frequency-selective fading, these long-range temporal dependencies and signal fluctuations constitute a major physical bottleneck, limiting transmission reliability.

Traditional adaptive equalization, spanning Linear Equalizer (LE) and Decision Feedback Equalizer (DFE), Least Mean Squares (LMS) and Recursive Least Squares (RLS) algorithms, has long been the standard for addressing signal distortion. Despite their established utility, these methods struggle with the deep ISI inherent in high-delay-spread channels. LMS-based schemes fail to maintain convergence under long-range interference, while RLS-based approaches suffer from prohibitive computational scaling and numerical risks.

In recent years, deep learning (DL) has revolutionized physical layer design by leveraging superior non-linear modeling and hierarchical feature extraction. Specifically, the end-to-end autoencoder framework in [1] established the theoretical foundation for DL-based communications. Subsequent research transitioned toward modular replacement, with architectures like DenseNet-based DeepReceiver in [2] and CNN-based schemes in [3] demonstrating high robustness against multipath fading. To enhance sequential modeling, [4] integrated Convolutional Neural Network (CNN) and Long Short-Term

Memory (LSTM) as CNN-LSTM structure for improved signal recovery. Recently, attention mechanisms have emerged as a focal point to combat long-range interference and high delay spread. Notably, [5] and [6] leveraged self-attention and sparse attention to enhance 5G reception, while Transformer-based pilot-free networks in [7] achieved robust equalization in time-varying channels. While the aforementioned methods demonstrate effective performance under low-to-moderate delay spread conditions, they fail to adequately address the extreme challenges posed by severe multipath propagation and high delay spread.

To bridge this gap, this paper proposes the Pilot-aided Feature-augmented Transformer Network (P-FTNet) for robust equalization under severe delay spreads. The primary contributions of this paper are twofold: First, we propose a novel pilot-aided conditioning mechanism that explicitly guides the equalization process, integrated with a multidimensional feature fusion strategy for robust representation learning. Second, extensive experiments under severe delay spreads demonstrate that the proposed P-FTNet yields a substantial performance gain, achieving a 4 dB improvement at an SER of 10^{-3} level over current state-of-the-art (SOTA) methods for 20-symbol delay spreads.

II. SYSTEM MODEL

A. Transmitted Signal Representation

We consider a single-input single-output (SISO) system employing single-carrier (SC) modulation. The source bit sequence $b[n]$ is channel-encoded and mapped onto a payload sequence $s[n]$ of length L via M -ary digital modulation. To acquire essential channel state information (CSI), a pilot sequence $p[n]$ of length L_p is concatenated with the payload, forming a transmission frame $z[n]$ with a total length of $L_f = L + L_p$.

B. Received Signal Representation

After pulse-shaping and transmission over a frequency-selective multipath fading channel, the discrete-time received signal $y[k]$ at the sampling level can be modeled as:

$$y[k] = \sum_{i=0}^{N-1} h_i[k]x[k - n_i] + w[k], \quad (1)$$

where $x[k]$ is the oversampled version of $z[n]$ after root-raised-cosine (RRC) filtering, and $h_i[k]$ denotes the complex gain of the i -th discrete channel tap with a corresponding sample delay n_i . N represents the total number of resolvable multipath

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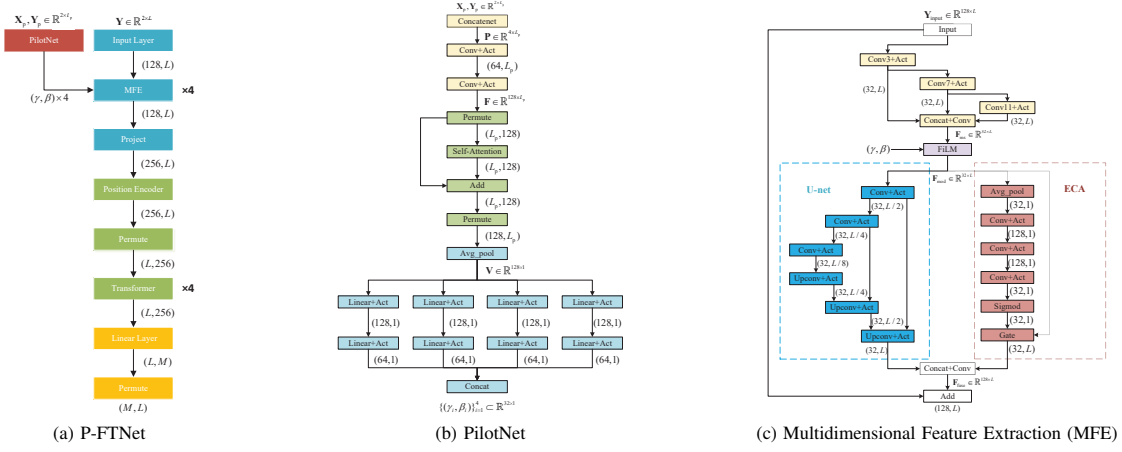


Fig. 1. The architecture of the proposed P-FTNet, along with detailed structures of the PilotNet and Multidimensional Feature Extraction (MFE) modules.

components, and $w[k]$ is the complex additive white Gaussian noise (AWGN).

At the receiver, the sequence $y[k]$ is processed by a matched filter and subsequently downsampled to the symbol rate. The resulting symbol-level sequence $\tilde{z}[n]$ is then fed into the equalization module to mitigate ISI and recover the transmitted symbols.

C. Problem Formulation

The core challenge in the considered system arises from the significant delay spread of the multipath fading channel. When the maximum delay spread $\tau_{max} = \max(\tau_i)$ is much larger than the symbol period T , the channel exhibits a long-tail impulse response, inducing severe and long-range inter-symbol interference (ISI).

To mitigate these distortions, our objective is to identify an optimal equalization mapping $\mathcal{F}^*$ belonging to a specific functional space $\mathcal{H}$ (e.g., the space of non-linear neural architectures or adaptive filtering structures), characterized by a parameter set Θ^* . This mapping aims to robustly recover the transmitted symbols $\hat{s}[n]$ from the corrupted observation sequence $\tilde{z}[n]$:

$$\hat{s}[n] = \mathcal{F}^*(z[n]; \Theta^*). \quad (2)$$

Upon obtaining the equalized symbols $\hat{s}[n]$, the sequence is processed by the demodulator to extract coded bits or soft-decision log-likelihood ratios (LLRs). Finally, the channel decoder performs error correction to reconstruct the estimated source bit sequence $\hat{b}[n]$.

III. P-FTNET: ADAPTIVE EQUALIZATION FOR SEVERE DELAY SPREAD CHANNELS

To effectively mitigate time-varying channel impairments while capturing complex signal dependencies, we propose the P-FTNet equalizer as $\mathcal{F}^*$ and find the optimal parameters Θ^* . The core design philosophy centers on a dual-path synergetic architecture: a conditioning branch (PilotNet) that extracts instantaneous channel descriptors, and a processing backbone that fuses multi-scale local features with global sequential context. The bridge between these paths is a Feature-wise Linear Modulation (FiLM) [8] mechanism, which enables the receiver to dynamically re-calibrate its feature extraction layers. In the following, we first delineate the end-to-end

data flow of P-FTNet, followed by a detailed elucidation of PilotNet's parameter generation process and multidimensional feature extraction (MFE) module.

A. Overall Architecture of P-FTNet

As illustrated in Fig. 1a, the architecture first employs a pilot network (PilotNet) to characterize the instantaneous channel state with known pilot $\mathbf{X}_p \in \mathbb{R}^{2 \times L_p}$ and received pilot $\mathbf{Y}_p \in \mathbb{R}^{2 \times L_p}$. This module generates scaling and shifting factors $\gamma, \beta \in \mathbb{R}^{32 \times 1}$ for FiLM in MFE module.

Simultaneously, the received data signal $\mathbf{Y} \in \mathbb{R}^{2 \times L}$ is projected into a 128-dimensional feature space $\mathbb{R}^{128 \times L}$ via an initial input layer. This is followed by four multidimensional feature extraction (MFE) modules, which are dynamically conditioned on the factors γ and β to adaptively refine local features based on specific channel impairments.

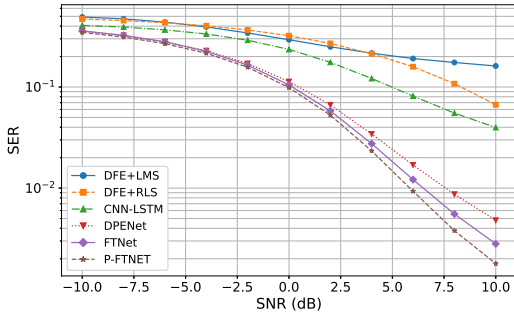
Subsequently, a projection layer expands the feature depth to $\mathbb{R}^{256 \times L}$. The resulting feature tensor is augmented with positional encoding and permuted to $\mathbb{R}^{L \times 256}$ to serve as the sequence input for a four-layer Transformer encoder [9]. The Transformer captures long-range dependencies, with the final equalization decisions yielded through a linear layer that maps the features back to the complex symbol space $\mathbb{R}^{M \times L}$.

B. Pilot Network (The Conditioning Branch)

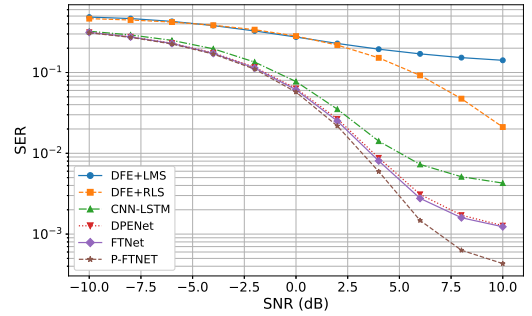
By concatenating the known pilot $\mathbf{X}_p$ with received pilot $\mathbf{Y}_p$ along the channel dimension, the network constructs a composite input tensor $\mathbf{P} \in \mathbb{R}^{4 \times L_p}$. A convolutional encoder, comprising two successive Conv+Act layers, transforms the input into a high-dimensional latent space.

Following the initial encoding, the refined features $\mathbf{F} \in \mathbb{R}^{128 \times L_p}$ are permuted to $\mathbb{R}^{L_p \times 128}$ to treat the pilot positions as a sequence for global contextual awareness via a Self-Attention mechanism [9]. The output of the attention mechanism is integrated with the original input $\mathbb{R}^{L_p \times 128}$ of the self-attention layer via a residual add operation to preserve primary channel features and permuted back to $\mathbb{R}^{128 \times L_p}$.

The resulting feature map is subsequently aggregated along the temporal dimension using global average pooling, yielding a global feature vector $\mathbf{V} \in \mathbb{R}^{128 \times 1}$. Then this global vector is passed through four parallel projection branches, compressing the 128-dimensional vector into a 32-dimensional representation. These decoupled heads independently compute the FiLM



(a) rayleigh channel



(b) rician channel

Fig. 2. The symbol error rate (SER) performance of different methods under (a) Rayleigh fading channel and (b) Rician fading channel.

parameter sets $\{(\gamma_i, \beta_i)\}_{i=1}^4 \subset \mathbb{R}^{32 \times 1}$, where each pair is used for layer-specific calibration of the subsequent feature extraction blocks.

C. Multidimensional Feature Extraction (MFE) Module

Initially, the input feature map $\mathbf{Y}_{\text{input}} \in \mathbb{R}^{128 \times L}$ undergoes a cascaded extraction process using three parallel convolutional branches with different kernel size. The resulting features from these branches are concatenated and fused via a convolution to form a unified multi-scale representation $\mathbf{F}_{\text{ms}} \in \mathbb{R}^{32 \times L}$.

Given the scaling factor γ_i and shifting factor $\beta_i \in \mathbb{R}^{32 \times 1}$ generated by the PilotNet (where i denotes the i -th MFE block), these parameters are expanded to match the 32 channels of the MFE module. The modulated feature $\mathbf{F}_{\text{mod}}$ is then calculated as:

$$\mathbf{F}_{\text{mod}} = (1 + \gamma_i) \odot \mathbf{F}_{\text{ms}} + \beta_i, \quad (3)$$

where $\odot$ denotes the element-wise multiplication across the channel dimension.

Building upon the modulated feature $\mathbf{F}_{\text{mod}}$, the architecture employs a dual-branch framework to concurrently refine data across the temporal and channel dimensions. The temporal branch utilizes a 1D U-net [10] for hierarchical encoding, while the channel branch adopts an efficient channel attention mechanism [11] to recalibrate the importance of different feature maps. Finally, the outputs from both branches are fused to produce the integrated feature map $\mathbf{F}_{\text{fuse}}$. To maintain structural integrity and enhance gradient flow, $\mathbf{F}_{\text{fuse}}$ is aggregated with the initial input through a global residual connection.

IV. SIMULATION RESULT

A. Experiment Setting

The P-FTNet is evaluated in a SISO-SC framework using BPSK modulation and a rate-1/2 LDPC code. Each frame consists of $L = 400$ payload symbols and $L_p = 60$ pilot symbols. Following 3GPP specifications [12], we employ Rayleigh (NLOS) and Rician (LOS) fading models with maximum delay spreads of $\{20, 30, 40\}$ and 20 symbols, respectively. To enhance generalization, path delays are randomly scaled within $[80\%, 120\%]$ during training. The architecture is implemented in PyTorch and optimized using AdamW with a cross-entropy loss function and a batch size of 256. The 30-epoch training procedure includes a 5-epoch linear warm-up followed by 25 epochs with an initial learning rate of 2×10^{-3} .

For a comprehensive performance evaluation, the proposed P-FTNet is benchmarked against a diverse set of baselines, ranging from DFE utilizing LMS and RLS algorithms as conventional techniques, to hybrid CNN-LSTM [4] and DPENet [6] as SOTA deep learning methods. To rigorously evaluate the individual contributions of the MFE and PilotNet, we developed and trained both the proposed P-FTNet and its variant FTNet, with the distinction being the presence or absence of the PilotNet. Based on the utilization of channel priors, the benchmark models CNN-LSTM, DPENet, and FTNet are categorized as non-pilot-aided methods. In contrast, DFE-LMS, DFE-RLS, and the proposed P-FTNet are pilot-aided architectures that actively leverage pilot information.

B. Performance

1) *Robustness to Diverse Channel Conditions:* Figure. 2a and Figure. 2b illustrate the symbol error rate (SER) performance of various algorithms under Rayleigh and Rician channels. The proposed FTNet and P-FTNet achieve the lowest SER across the entire SNR range. Specifically, traditional techniques such as DFE-LMS and DFE-RLS suffer from performance bottlenecks, with the former exhibiting a noticeable error floor at high SNRs. While the CNN-LSTM and DPENet benchmarks offer substantial improvements over these conventional methods, they still fall short of our proposed architectures. Among all evaluated schemes, P-FTNet consistently delivers the superior performance. At an SER of 10^{-3} level, the proposed P-FTNet achieves a performance gain of 2.5 dB and 4 dB over the DPENet in Rayleigh and Rician channels respectively. This underscores the efficacy of integrating pilot-aided refinement, multidimensional feature extraction, and self-attention mechanisms, which collectively ensure robust channel equalization in complex fading environments. In addition, the consistent gap between P-FTNet and FTNet quantifies the performance uplift provided by the PilotNet-based conditioning, confirming that explicit channel priors significantly sharpen the decision boundaries of MFE.

2) *Resilience to Multipath Delay Spread:* To evaluate the generalization capability of the proposed models under channel mismatch conditions, the deep learning-based architectures were trained under a channel condition with a maximum delay spread of 20 symbols and subsequently tested across varying delay spreads at a fixed SNR of 10dB. As illustrated in Figure. 3, the proposed P-FTNet and its variant FTNet consistently attain superior performance across all tested configurations. In regimes with relatively small delay spreads,

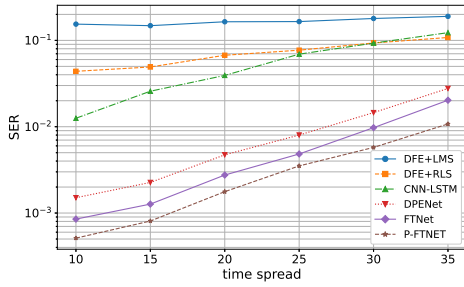


Fig. 3. The performance of different methods under different max time spread of rayleigh channel.

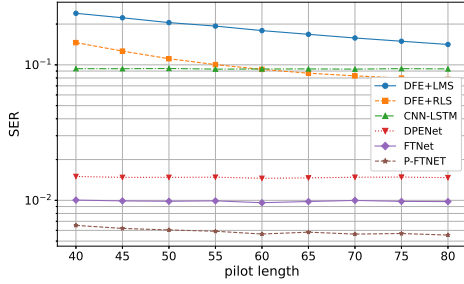


Fig. 4. The impact of pilot length on different methods.

deep learning-based methods exhibit a significant advantage over conventional equalization algorithms. However, as the delay spread increases, owing to its ability to leverage pilot information, the DFE-RLS algorithm surpasses the CNN-LSTM model. Notably, architectures utilizing Transformer-based backbones (P-FTNet, FTNet, and DPENet) maintain a significant performance margin, showcasing their efficacy in modeling long-range ISI. Among these, P-FTNet preserves its lead over FTNet and DPENet, further validating the importance of its pilot-aided conditioning branch in providing instantaneous channel awareness.

3) *Spectral Efficiency and Pilot Sensitivity*: Fig. 4 illustrates the performance trajectory of various equalizers as a function of pilot length. To rigorously test the models, they were trained under 20-symbol delay spread and evaluated under a mismatched, more severe 30-symbol condition. For deep learning-based benchmarks such as CNN-LSTM, DPENet, and FTNet, their SER performance is virtually invariant to pilot length since these architectures do not utilize pilot during inference. In contrast, the performance of DFE-LMS, DFE-RLS, and P-FTNet improves as the pilot length increases, with P-FTNet maintaining a consistent and significant advantage across all configurations. Crucially, P-FTNet achieves a high level of reliability even with a minimal pilot length of 40. This represents a low training overhead of only 9.09% relative to the total frame length, outperforming all competitors and thereby maximizing spectral efficiency under resource-constrained scenarios.

4) *Complexity and Performance Trade-off*: Regarding the analysis of model complexity and parameter efficiency, we focus on the four deep learning-based architectures, as the performance of conventional methods under high-delay spread conditions is insufficient for practical requirements. Among these, the proposed P-FTNet achieves the most superior performance while maintaining a highly competitive overhead. Detailed metrics are summarized in Table I. Specifically, the

TABLE I
MODEL COMPLEXITY AND PERFORMANCE COMPARISON.

Method	Parameters	Flops
CNN-LSTM	1.407M	0.194G
DPENet	1.138M	0.402G
FTNet	0.893M	0.300G
P-FTNet	1.117M	0.308G

number of parameters for P-FTNet is 1.117M, which is lower than that of CNN-LSTM and DPENet, while representing a moderate increase over its variant FTNet. In terms of computational complexity, the FLOPs of P-FTNet are recorded at 0.308G. This value represents a marginal increment compared to FTNet and is significantly lower than that of DPENet. Although it is higher than the CNN-LSTM model, the substantial gain in equalization accuracy justifies this additional cost.

V. CONCLUSION

This paper proposes a P-FTNet equalizer to address the challenges of severe ISI and long-range temporal dependencies in dense urban wireless channels. By integrating PilotNet with MFE, the proposed architecture effectively leverages limited reference signals to enhance the global sequence modeling capabilities of the Transformer. Experimental results under different channel conditions demonstrate the proposed P-FTNet significantly outperforms traditional adaptive equalizers and SOTA deep learning benchmarks.

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